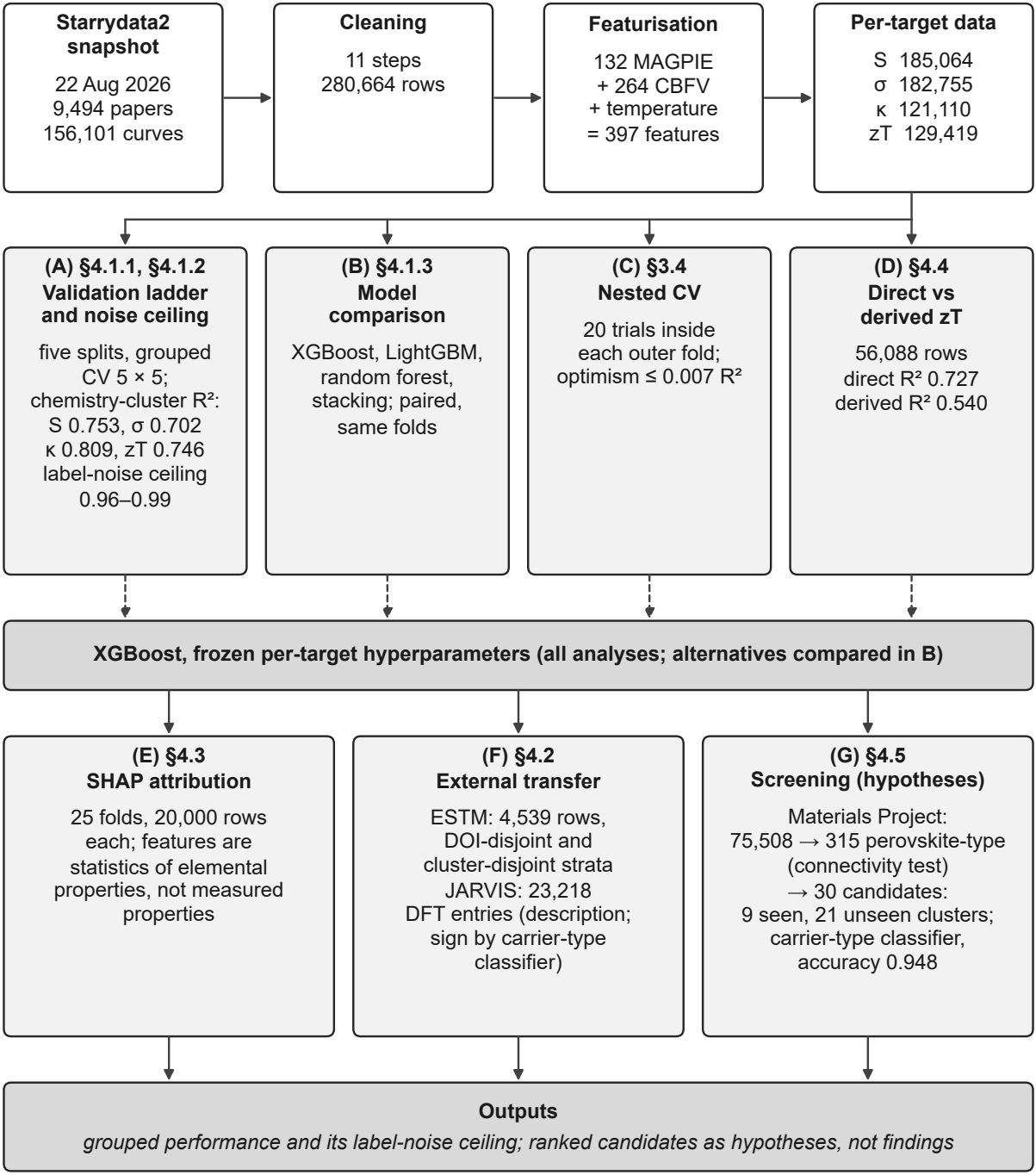

Starrydata2 snapshot
22 Aug 2026
9,494 papers
156,101 curves
Cleaning
11 steps
280,664 rows
Featurisation
132 MAGPIE
+ 264 CBFV
+ temperature
= 397 features
Per-target data
S 185,064
σ 182,755
κ 121,110
zT 129,419
(A) §4.1.1, §4.1.2
Validation ladder and noise ceiling
five splits, grouped CV 5 × 5;
chemistry-cluster R²:
S 0.753, σ 0.702
κ 0.809, zT 0.746
label-noise ceiling 0.96–0.99
(B) §4.1.3
Model comparison
XGBoost, LightGBM, random forest, stacking; paired, same folds
(C) §3.4
Nested CV
20 trials inside each outer fold; optimism ≤ 0.007 R²
(D) §4.4
Direct vs derived zT
56,088 rows
direct R² 0.727
derived R² 0.540
XGBoost, frozen per-target hyperparameters (all analyses; alternatives compared in B)
(E) §4.3
SHAP attribution
25 folds, 20,000 rows each; features are statistics of elemental properties, not measured properties
(F) §4.2
External transfer
ESTM: 4,539 rows, DOI-disjoint and cluster-disjoint strata
JARVIS: 23,218 DFT entries (description; sign by carrier-type classifier)
(G) §4.5
Screening (hypotheses)
Materials Project:
75,508 → 315 perovskite-type (connectivity test)
→ 30 candidates:
9 seen, 21 unseen clusters;
carrier-type classifier, accuracy 0.948
Outputs
grouped performance and its label-noise ceiling; ranked candidates as hypotheses, not findings